Dynamic Entropic Tail-State Estimation under Asymmetric Gamma Jumps

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Abstract

The Giorno–Tsallis research program investigates how distributional tail risk may be inferred from observations and subsequently incorporated into optimal economic decisions. The present work develops its econometric foundation, distinguishing the non-extensivity parameter q from an entropy functional and estimating latent tail-regime beliefs in a two-state q-Gaussian (Student-t) hidden Markov model. A nine-parameter extension adds independent reflected-Gamma jumps. We establish that the resulting asymmetry cannot generally be reproduced by a common location shift in a symmetric Student-t HMM, without claiming full nine-parameter global identification. The empirical design comprises 2,700 matched paths covering clean Student-t observations, Gaussian contamination, and negative Gamma jumps, with training sizes 1,000, 2,000 and 3,000 and 400 held-out observations. Under negative Gamma jumps at n=3,000, HMM-9 lowers q1 estimation RMSE from 0.1978 (HMM-6) and 0.1976 (location-adjusted HMM-7) to 0.0663 and improves mean predictive log score by 0.01830 per observation relative to HMM-7 (paired 95% Monte Carlo interval [0.01712, 0.01948]). The gain is not universal in clean data or symmetric contamination. Separate nine-parameter inference simulations reveal boundary and joint Wald-coverage limitations. The estimator delivers filtered beliefs about latent tail risk and asymmetric jumps; economic welfare effects are not evaluated here.[1] Dynamic Gamma shape, structural breaks, and domain-specific physical applications remain directions for further investigation.

Keywords: Tsallis entropy; q-Gaussian; Student-t; hidden Markov model; Gamma jumps; density forecasting; tail-state filtering; likelihood inference; Monte Carlo.

JEL classification (provisional): C13, C15, C32, C58, G17.

1. Introduction

Financial return dynamics display varying tail thickness, temporal dependence, and abrupt extreme observations. These features may be confounded in a conventional latent-regime specification: a large return could reflect an ordinary draw from a heavy-tailed regime, an increase in the scale parameter, or a separate shock mechanism. Distinguishing these explanations is important for the estimation of latent tail states, forecasting conditional densities, and the validity of parameter uncertainty statements.

The present article constitutes the statistical component of an originally unified Giorno–Tsallis program. Optimal quoting and the economic value of information under filtered beliefs about nonextensive tail risk remain the subjects of a separate, ongoing paper. Dividing the research into two articles isolates statistical inference from economic choice without reducing the original objective to a competition between algorithms.

The central research question is: under what circumstances can a filtered entropic tail index be interpreted as evidence about structural tail risk, rather than as an artifact of unmodelled asymmetric shocks, and what statistical object is consequently available to an economic decision-maker?

The present work develops and evaluates a two-state q-Gaussian hidden Markov framework. The original six parameters describe Markov persistence, tail shape, and conditional scale. We then introduce an explicit reflected-Gamma jump component with three additional parameters, producing a nine-dimensional likelihood. The purpose is not to relabel a pre-existing mixture model as a new probability family, but to test a precise econometric proposition: a correctly specified asymmetric shock mechanism can prevent exceptional returns from being incorrectly absorbed into estimated entropic tail thickness.

[1] Borak, S., Misiorek, A., and Weron, R. (2011) and Chakraborty, S., Nanda, A. K., and Balakrishnan, N. (2026).

Our contribution consists of four elements. First, we formalize a q-Gaussian latent tail-state model with a normalized asymmetric Gamma-jump convolution, and distinguish the estimated non-extensive parameter from an entropy functional. Second, we prove an observable model-family distinction: a nondegenerate reflected-Gamma jump cannot be reproduced by a common location shift of symmetric Student-t HMM emissions. Third, we evaluate matched HMM-6, location-adjusted HMM-7 and HMM-9 estimators against known generators with 2,700 paths and chronological test windows. Fourth, we report separate HMM-9 Wald limitations rather than treating forecast accuracy as proof of global identification or nominal inference.[2]

Recent evidence on asymmetric reversals in stock returns (Baldi-Lanfranchi, Collin-Dufresne, and Daniel, 2026) motivates the investigation of sign-dependent disturbances; it does not, however, provide an identification proof for this model. Neither the presence of fat tails nor superior predictive density scores establish the validity of a particular estimation theorem.

2. Literature review and theoretical positioning

2.1 Nonextensive statistical mechanics and the meaning of q

Tsallis (1988) introduced an entropy functional that generalizes Boltzmann–Gibbs statistics, and Tsallis (2009) develops its mathematical and physical setting. The associated q-Gaussian family provides one way of representing departures from Gaussian decay, while the generalized central-limit framework of Umarov, Tsallis, and Steinberg (2008) studies the appearance of q-Gaussian attractors under special dependence structures. These are foundational results, but they do not by themselves identify an evolving q from a noisy, partially observed time series. Nor does observing a Student-t-like marginal distribution establish the microscopic mechanism responsible for nonextensivity.

For the parameter range examined here, q-Gaussian densities can be written exactly in Student-t form using a monotone transformation between q and the degrees of freedom. This equivalence supplies an estimable statistical observation law; it does not imply that the time-varying posterior mean of q equals a measured Tsallis entropy. The latter is a functional of a normalized distribution and depends on the entropy order, reference measure, and treatment of mixtures. Our contribution is located at this statistical interface: inference on latent distributional tail states when asymmetric disturbances are present. The thermodynamic interpretation of such estimates in a particular physical system requires additional model-specific justification.

2.2 Heavy tails, asymmetry, and conditional volatility in financial econometrics

Cont (2001) documents the empirical regularities that motivate non-Gaussian descriptions of asset returns, including pronounced tails and volatility clustering. Broad treatments by Jondeau, Poon, and Rockinger (2007) and Borak, Misiorek, and Weron (2011) place Student-t and other heavy-tailed distributions within a wider set of financial modeling approaches. Importantly, heavy tails can reflect more than one mechanism: persistent latent regimes, changing conditional scale, skewness, and occasional jumps may all contribute to unusually large observations. A fitted tail index can therefore be sensitive to the assumed mechanism for the rest of the conditional distribution.

Fernández and Steel (1998) provide a flexible treatment of fat tails and skewness, while Giot and Laurent (2003) examine asymmetric risk modeling for long and short positions. Chib, Nardari, and Shephard (2002) establish the relevance of latent stochastic-volatility models and their likelihood-based Bayesian treatment. These alternatives clarify the scope of our comparison: an asymmetric Gamma-jump HMM is not claimed to dominate skew-t, asymmetric volatility, or stochastic-volatility specifications universally. Instead, the paper asks a narrower question: when the data contain an independently occurring, negatively signed shock, can an explicit jump convolution help distinguish that shock from the latent thickness of ordinary tails?

[2] Douc, R., Moulines, E., and Rydén, T. (2004).

2.3 Markov switching, likelihood inference, and nonregularity

Hamilton (1989) established Markov switching as a useful representation of shifts in economic dynamics. For hidden Markov models, Rydén (1994), Bickel, Ritov, and Rydén (1998), and Douc, Moulines, and Rydén (2004) develop consistency and asymptotic-normality results under specific identification, dependence, and regularity conditions. McLachlan and Peel (2000) discuss closely related estimation and identification issues in finite mixtures. The attraction of the HMM formulation is that the observation likelihood can be evaluated through a finite-state forward recursion while persistence is represented through transition probabilities.

These general statistical results cannot simply be transferred to every boundary of the proposed nine-parameter model. At zero jump probability, the Gamma shape and scale disappear from the observable law, so nuisance parameters are unidentified under the no-jump null.[3] Hansen (1992) provides an influential reminder that likelihood-ratio testing in regime-switching settings can be nonstandard. Consequently, routine chi-square reference distributions and full-parameter Gaussian Wald statements require scrutiny, even if numerical maximization succeeds. The present paper separates exact emission and filtering identities from model-specific global identification and from finite-sample confidence-interval coverage.

2.4 Gamma-based shocks and alternative jump mechanisms

The jump literature contains several distinct ways of generating asymmetric return distributions. Madan, Carr, and Chang (1998) construct the Variance Gamma process by subordinating Brownian motion with a Gamma process; Carr, Geman, Madan, and Yor (2002) examine the fine structure of observed returns. Kou (2002) introduces a double-exponential jump-diffusion model, while Eraker, Johannes, and Polson (2003) and Maheu and McCurdy (2004) study jumps in returns, volatility, and news-related dynamics. These studies motivate allowing discontinuous, potentially asymmetric shocks rather than explaining every extreme observation through a heavier continuous tail.

Our reflected-Gamma specification is not a Variance Gamma process and is not a Gamma time-change of Brownian motion. It combines a Bernoulli jump indicator with a nonnegative Gamma magnitude, subtracting the magnitude from an otherwise symmetric regime-dependent Student-t innovation. This construction produces a normalized one-sided shock convolution with parameters for jump incidence, shape, and scale. It provides a controlled setting for studying the statistical confounding between ordinary heavy-tail regimes and exceptional asymmetric shocks; it does not establish that such shocks are uniquely Gamma distributed in observed financial or physical data.

2.5 Filtered risk beliefs and economic decisions

The relevance of conditional tail risk for economic decisions motivates but does not determine the statistical model. Avellaneda and Stoikov (2008) develop an optimal-quoting framework in which inventory risk and execution intensities jointly affect liquidity-provision decisions. Recent evidence by Baldi-Lanfranchi, Collin-Dufresne, and Daniel (2026) documents asymmetry in short-horizon return reversals; that result motivates attention to the sign of shocks but does not validate our particular Gamma convolution. In the broader Giorno–Tsallis research program, the output of the current paper is a filtered distribution over latent risk states and a jump posterior. A separate optimal-control analysis must determine whether those beliefs change the optimal policy or its welfare performance.

2.6 Research gap, contribution, and possible relevance beyond finance

Taken together, these literatures offer well-established ingredients—nonextensive distribution families, state-space inference, asymmetric return laws, and jump models—but do not automatically answer the specific identification question posed here. The paper contributes a transparent two-state q-Gaussian model with a shared reflected-Gamma mechanism, an exact finite-state filtering construction, and a proof that its nondegenerate asymmetric marginal law cannot be replicated by a common location shift of a symmetric Student-t HMM.

[3] Carr, P., Geman, H., Madan, D. B., and Yor, M. (2002), Douc, R., Moulines, E., and Rydén, T. (2004) and Eraker, B., Johannes, M., and Polson, N. (2003).

Matched comparisons against HMM-6 and location-adjusted HMM-7 then measure parameter recovery and predictive performance under both correctly specified and deliberately misspecified generators. The contribution is this model-specific separation and its assessed limits, rather than the invention of q-Gaussians, hidden Markov models, or Gamma mixtures.

The same distinction may matter in physical time series exhibiting persistent background fluctuations and intermittent sign-asymmetric bursts—for example, when an instrument records a symmetric heavy-tailed baseline plus occasional directional impulses. Applying the framework outside finance would require defining the physical observation process, identifying the appropriate sign and scale of disturbances, and comparing the model with physical alternatives. It would not follow from the present Monte Carlo alone that any system obeys a generalized central-limit mechanism or that its thermodynamic entropy has been estimated. This cross-disciplinary possibility is therefore offered as a testable modeling perspective, not an established physical law.

2.7 Cross-disciplinary work on estimating q and the entropy functional

The empirical determination of a non-extensive index must be distinguished from the estimation of an entropy functional at a specified order. Plastino and Plastino (1999) place Tsallis entropy within the Jaynes maximum-information framework, while Lavenda and Dunning-Davies (2005) scrutinize pseudo-additivity and its interpretation. These studies concern the mathematical and physical meaning of generalized entropy; neither supplies a universal time-series estimator for a latent q. Moreover, the entropic order a used to evaluate S_a[h] need not be identified with a fitted q-Gaussian tail index q_j. Treating these objects as interchangeable would obscure the purpose of the present estimator.

Chakraborty, Nanda, and Balakrishnan (2026) investigate estimation of Tsallis entropy itself, proposing estimators based on higher-order sample spacings and quantiles, with goodness-of-fit applications and theoretical properties for a quantile-based estimator. In this setting the entropy order is an input to the estimation problem. By contrast, our HMM estimates a distributional tail index and filtering probabilities within a specified parametric observation family. The two approaches can be compared through a common entropy functional evaluated from the predictive distribution, but they are not estimates of the same quantity by definition.

Quantum-information research illustrates a third distinction. Bao, Gao, and Wang (2026) study the sample and query complexity of estimating quantum Tsallis relative entropy between quantum states for fixed $\alpha \in (0,1)$; they do not infer a dynamically changing classical q-Gaussian tail index. Wu et al. (2026), in test-time adaptation of vision–language models, construct class-specific values of q from estimated prediction bias; Kanzawa (2020) uses Tsallis entropy in fuzzy latent semantic optimization. Such q-dependent computational procedures show the breadth of the entropy formalism, but their learned or selected indices do not automatically have a thermodynamic interpretation.

2.8 Physical interpretations and transferability of latent q-state inference

In physical applications, the meaning of q depends on the observable, the proposed ensemble or dynamics, and the distribution fitted. Sigalotti, Ramírez-Rojas, and Vargas (2023) review empirical q-indices for earthquake magnitudes, inter-event times and distances, including time-varying seismic estimates. Their q_M, q_T and q_D arise from distinct q-statistical models and should not be numerically equated with the q obtained here from Student-t degrees of freedom. Masthay, Benton, Guerrieri, and Keil (2024) relate an index q to deviations from global thermal equilibrium in their paramagnetic spin-lattice construction; the resulting interpretation depends on that physical model and is not a general theorem about financial returns.

The potential cross-disciplinary contribution of the present paper is therefore methodological and testable: a latent heavy-tailed background may be confounded with intermittent, sign-asymmetric disturbances, and an explicit shock component may improve inference on the background distribution when that model is suitable. Physical transfer requires a justified observation law, dimensionless scaling, appropriate time-resolution, an assessment of alternative noise and impulse mechanisms, and external physical validation. The wider objective of

a unified entropy theory spanning physical and economic behavior remains a motivating research vision rather than a conclusion established by HMM-9.

3. Baseline non-extensive HMM

3.1 q-Gaussian and Student-t parameterization

Let $K_t \in \{0,1\}$ be a latent Markov state. Conditional on $K_t = j$, the observation r_t is drawn from a centered Student-t distribution with degrees of freedom $\nu_j > 2$ and scale $s_j > 0$. The q-Gaussian shape index q_j is obtained as a one-to-one transformation of ν_j over the maintained parameter range:

$$\nu_j = \frac{3 - q_j}{q_j - 1}, \quad q_j = \frac{\nu_j + 3}{\nu_j + 1}, \quad 1 < q_j < 5/3 \tag{1}$$

$$f_j(r) = \frac{\Gamma(\frac{\nu_j + 1}{2})}{s_j \sqrt{\pi \nu_j}\, \Gamma(\frac{\nu_j}{2})} \cdot [1 + \frac{r^2}{\nu_j s_j^2}]^{-\frac{\nu_j + 1}{2}} \tag{2}$$

Here, s_j denotes the Student-t scale, not necessarily its conditional standard deviation. The restriction $q_j < 5/3$ guarantees finite conditional variance. The symmetric emission density itself exhibits no directional skewness. This model represents a discrete-time observation distribution, not an assertion that Brownian increments follow q-Gaussian laws.

3.2 Markov transitions and the six parameters

$$P = [[p_{00}, 1-p_{00}], [1-p_{11}, p_{11}]], \quad \theta_6 = (p_{00}, p_{11}, q_0, q_1, s_0, s_1). \tag{3}$$

We impose $q_0 < q_1$ to fix regime labels and assume interior, ergodic transitions for regular estimation. These six parameters are jointly estimated from the observed return sequence via likelihood maximization. Persistent regime probabilities generate a dynamic filtered q index, even though q_0 and q_1 remain static within their respective regimes.

3.3 The distinction between non-extensivity, entropy, and economic risk

The Tsallis (1988,2009) index q is a distribution-shape parameter; it is not itself a value of entropy. To avoid confusing q with entropy or treating them as the same concept, we define a standardized, dimensionless variable x and a normalized predictive or regime density $h(x)$. For a fixed reference order $a > 0$, $a \neq 1$, the differential Tsallis functional can be written as $S_a[h] = \frac{1 - \int h(x)^a \, dx}{a - 1}$, provided the integral exists. In the limit $a \to 1$, $S_a[h]$ approaches Shannon differential entropy under suitable integrability. The standardization and choice of a must be declared before entropy values are compared across scales or time.

The filtered index $\hat{q}_t|_t = \sum_j \alpha_t|_t(j)\hat{q}_j$ summarizes posterior uncertainty about the latent tail regime. In general, the mixture predictive density $\sum_j \alpha_t|_{t-1}(j) g_j(r)$ is not itself q-Gaussian and its Tsallis entropy is not obtained by plugging $\hat{q}_t$ into a regime-specific formula. These are different statistical objects. This manuscript directly estimates q, jump parameters and the posterior regime probabilities; estimation of the full predictive-entropy functional is a defined, possible extension rather than a completed finding.

This distinction is essential for the economic interpretation. Whether that information changes optimal quotations depends separately on the chosen utility, execution technology and control-state dynamics.

4. The nine-parameter asymmetric Gamma-jump specification

Let B_t be an independent Bernoulli jump indicator and G_t be a Gamma-distributed jump magnitude. The return innovation is

$$r_t = s_{K_t} T_{\nu_{K_t},t} - B_t G_t, \quad \mathrm{B_t} \sim Bernoulli(\varepsilon), \quad G_t \sim Gamma(k, \text{scale } \theta). \tag{4}$$

The reflected Gamma variable $-G_t$ is nonpositive; the shape k and scale θ are strictly positive. Jump parameters are shared across HMM states in the current experiment and independent of K_t. The joint nine-parameter vector is

$$\Theta_9 = (p_{00}, p_{11}, q_0, q_1, s_0, s_1, \varepsilon, k, \theta). \tag{5}$$

Conditional on state j, integrating out the unobserved jump occurrence and jump magnitude gives the normalized observation density:

$$g_j(r; \Theta_9) = (1-\varepsilon) f_j(r) + \varepsilon \int_0^\infty \frac{f_j(r+z)\, z^{k-1}\, exp(-z/\theta)}{\Gamma(k)\, \theta^k}\, dz \tag{6}$$

Because the Gamma kernel integrates to one and f_j is a density, the convolution is a proper density, and so is the convex mixture. This property permits likelihood-based estimation. It is not equivalent to the merely algorithmic Gamma-precision representation of the ordinary Student-t distribution.

4.1 A distinct, exact Gamma-precision identity

$$\Lambda_t \mid K_t = j \sim Gamma(\frac{\nu_j}{2}, rate\ \frac{\nu_j}{2}). \tag{7}$$

Integrating out Λ_t gives precisely equation (2). Therefore, Gamma-precision augmentation changes an estimation algorithm but not the six-parameter observed-data likelihood. By contrast, the negative-jump mechanism in equation (6) genuinely changes the observational model. The previous Gamma-EM comparison using 3,000 datasets found the direct and polished augmented MLEs essentially equivalent, as expected under this identity.

4.2 The no-jump boundary

At $\varepsilon = 0$, the extra shape and scale parameters (k, θ) do not affect the observed density and are not identified. Consequently, ordinary nine-parameter Wald intervals and conventional χ^2 likelihood-ratio tests for a jump component are not valid at this boundary without nonstandard adjustments. The numerical optimizer used in the reported experiment constrained $\varepsilon \geq 0.00001$; an estimate close to this lower limit is only an approximation to a zero-jump fit.

4.3 An observable theoretical distinction: asymmetric jumps versus location shifts

Consider the seven-parameter location-adjusted benchmark $r_t = \mu + s_{K_t} T_{\nu_{K_t},t}$, with a common unknown μ and the same two-state Markov transitions. At every time t, its unconditional marginal distribution is symmetric about μ because each state-conditional Student-t density is symmetric around zero. In contrast, the HMM-9 observation is $r_t = s_{K_t} T_{\nu_{K_t},t} - B_t G_t$, where B_t is an independent Bernoulli(ε) and G_t is an independent $Gamma(k, \theta)$.

Proposition 3 - non-equivalence to a common location shift.
Let every regime-conditional Student-t component be centered with positive scale and finite positive degrees of

freedom. Suppose the reflected-Gamma jump is independent of the latent state and Student-t innovations, $\varepsilon > 0$, $k > 0$ and $\theta > 0$. Then the one-time marginal law of the reflected-Gamma-jump model is not symmetric about any finite location, and consequently cannot equal the marginal law of any two-state Student-t HMM whose regime densities share a common location μ.[4] This result separates the model families, but does not establish global identification of all nine parameters.

Proof sketch. A centered Student-t law is a positive Gaussian-scale mixture and therefore has an everywhere strictly positive, real, even characteristic function. Any mixture across latent regimes preserves that property. If the convolution of this symmetric base with $J = -BG$ were symmetric about a constant c, the positivity of the base characteristic function would force J to be symmetric about c. Yet J has support bounded above by zero, is unbounded below, and has a nonzero left tail, which is impossible for a distribution symmetric about a finite c. A full characteristic-function argument is given in Appendix A.5.

5. Exact filtering and dynamic tail-index estimation

Write $\alpha_t(j) = Pr(K_t=j \mid r_1,\ldots,r_t)$. For a fixed parameter Θ and initial state probability vector, the finite-state Bayesian filter consists of prediction followed by normalization:

$$\alpha_{t|t-1}(j) \;=\; \sum_i \; \alpha_{t-1|t-1}(i)\, P_{ij} \text{ and } \; \alpha_{t|t}(j) \;=\; \frac{g_j(r_t)\, \alpha_{t|t-1}(j)}{\sum_l g_l(r_t)\, \alpha_{t|t-1}(l)} \tag{8}$$

$$\hat{q}_{t|t} \;=\; \hat{q}_0\, \alpha_{t|t}(0) \;+\; \hat{q}_1\, \alpha_{t|t}(1) \tag{9}$$

Equation (8) is an exact finite-state filter conditional on specified parameters and the assumed observation law. It is not a proof that a misspecified HMM or a numerical approximate filter recovers the true state path with vanishing error. In the Monte Carlo experiments the filtered q estimate is evaluated against known simulated latent states, but those latent labels are not supplied to the fitting algorithm.

The model also implies a conditional posterior jump probability. Writing $h_j(r)$ for the convolution term in equation (6),

$$Pr(B_t = 1 \mid r_{1:t}) \;=\; \sum_j \; \alpha_{t|t}(j) \;\cdot\; \frac{\varepsilon\, h_j(r_t)}{g_j(r_t)}. \tag{10}$$

This posterior probability differs conceptually from an Isolation Forest anomaly flag. A rare heavy-tailed observation can receive a low posterior jump probability when well explained by an existing HMM regime.

5.1 The statistical object handed to the economic theory

For estimated structural parameter vector $\hat{\theta}_9$, the observation history $r_{1:t}$ yields a plug-in posterior $\hat{\alpha}_{t|t} \;=\; (\widehat{Pr}(K_t = 0 \mid r_{1:t}), \widehat{Pr}(K_t = 1 \mid r_{1:t}))$. The predictive density for the next observation is $\hat{p}(r_{t+1} \mid r_{1:t}) \;=\; \sum_j \; \hat{\alpha}_{t+1|t}(j)\, g_j(r_{t+1};\, \hat{\theta}_9)$. An optimizer facing partial information should treat this conditional predictive law or the posterior belief (and, when warranted, parameter uncertainty) as the statistical state. Replacing the belief by its scalar mean $\hat{q}_t$ is an approximation, not an identity.

The conditional probability of an asymmetric jump provided by equation (10) is a second, conceptually separate risk summary. A market-making model can use both the posterior tail-regime probability and the jump

[4] Carr, P., Geman, H., Madan, D. B., and Yor, M. (2002), Donnelly, R., and Jaimungal, S. (2024) and Douc, R., Moulines, E., and Rydén, T. (2004).

probability, but any implied improvement in optimal utility must be verified within a future decision model.[5] Neither lower filtering RMSE nor a higher density score alone proves a welfare gain.

6. Identification, regularity, and inferential boundaries

Proposition 1. Identification within the centered q-Gaussian emission family

For $\nu>2$ and $s>0$, the normalised centred Student-t density in (2) is identifiable in (ν,s), and hence in (q,s) under the one-to-one transformation (1).

Proof: if two such densities agree almost everywhere, their polynomial tail exponents must agree, implying equal ν. Conditional on equal ν, equality of the density at zero, which is proportional to $1/s$, implies equal scale s. This establishes emission-family injectivity, not unconditional global identification of the two-state mixture with nine unknown parameters.

Proposition 2. Filtering identity and positivity qualifications

Given a specified transition matrix, initial state probabilities, and normalised observation densities g_j that are positive on the observation support, equation (8) follows from the Chapman–Kolmogorov prediction equation and Bayes' rule. In a strictly positive finite-state transition model, certain forms of filter contraction may hold under additional uniform likelihood-ratio or mixing conditions. Because Student-t observations are unbounded, no uniform bounded-observation filter-error theorem is asserted here without a separate proof.

Conditional asymptotic-inference statement

For a stationary, identifiable HMM with an interior true parameter, an appropriate domination and differentiability envelope, sufficiently mixing hidden dynamics, nonsingular Fisher information, and a global observed-data likelihood maximiser, the standard HMM likelihood framework supports consistency and asymptotic normality. Those are conditional generic results, not yet a model-specific verification for the full reflected-Gamma nine-parameter family. In particular, near $\varepsilon = 0$ the information matrix degenerates in jump nuisance directions, and a Gaussian contamination generator is not the correctly specified nine-parameter model. Therefore, claiming nominal nine-parameter Wald coverage would currently be premature.[6]

7. Likelihood computation and experimental design

The HMM-6 likelihood is evaluated by the standard forward recursion with centred Student-t emissions. The HMM-9 convolution in equation (6) is computed by FFT-based discretisation of the Gamma-shock mixture, using a return grid approximately [−250,250] and step 0.025, with Gamma bin probabilities and interpolation. For the reviewed implementation, a quadrature comparison found a maximum absolute log-emission discrepancy of roughly 0.003589 over its audit grid.[7]

Both models are fitted by bounded L-BFGS-B from multiple starting values, with labels ordered by q. The Gamma search bounds are $\varepsilon \in [0.00001,0.15]$, $k \in [0.55,7]$, $\theta \in [0.2,8]$. In each scenario, n and replication ID, the same return path is given to HMM-6 and HMM-9; simulated hidden states and jump flags are not supplied to the fitting algorithms. The other exercise is a completed 300-replication-per-cell experiment, not a continuation or additive pooling of the older 500-replication benchmark.

Numerical sensitivity remains a substantive qualification: the exact convolution likelihood and its FFT approximation need not yield identical finite-sample estimates. A finer-grid and quadrature sensitivity

[5] Borak, S., Misiorek, A., and Weron, R. (2011) and Chakraborty, S., Nanda, A. K., and Balakrishnan, N. (2026).

[6] Hansen, B. E. (1992).

[7] Borak, S., Misiorek, A., and Weron, R. (2011), Chib, S., Nardari, F., and Shephard, N. (2002), Cont, R. (2001) and Fernández, C., and Steel, M. F. J. (1998).

assessment, including investigation of the six HMM-9 optimizer non-success flags, remains important for replication and subsequent inference. The 2,700-path forecasting results establish performance under the stated numerical implementation, not a generic bound on approximation error.

Generator	Training n	Verified paths	Held-out n
Clean Student-t	1000	300	400
Clean Student-t	2000	300	400
Clean Student-t	3000	300	400
Gaussian contamination	1000	300	400
Gaussian contamination	2000	300	400
Gaussian contamination	3000	300	400
Negative Gamma jumps	1000	300	400
Negative Gamma jumps	2000	300	400
Negative Gamma jumps	3000	300	400

Table 1. Completed matched MOON design (13 October 2026). Nine cells contain 300 independent simulated paths apiece (2,700 total), each with 400 chronological held-out observations.

The generators use $p_{00}=0.985$, $p_{11}=0.960$, $q_0=1.12$, $q_1=1.42$, $s_0=0.80$, $s_1=1.25$. Clean data have no additional shock. Gaussian-contaminated data add independent $N(0,5^2)$ noise with probability 0.03. Negative-Gamma-jump data subtract Gamma(shape 2, scale 3) with probability 0.03. The first case is a no-jump boundary under HMM-9; the second is misspecified under the negative-Gamma model; the third matches the HMM-9 jump family in the interior.

8. Main Monte Carlo findings

8.1 Recovery of heavy-tail and scale-related parameters

Generator	n	Reps	q_1 RMSE H6	q_1 RMSE H9	q_0 RMSE H6	q_0 RMSE H9
Clean Student-t	1000	300	0.0954	0.1088	0.0702	0.0693
Clean Student-t	2000	300	0.0614	0.0651	0.0533	0.0530
Clean Student-t	3000	300	0.0519	0.0543	0.0434	0.0440
Gaussian contamination	1000	300	0.1465	0.1254	0.1432	0.1370
Gaussian contamination	2000	300	0.1553	0.1279	0.1209	0.1183
Gaussian contamination	3000	300	0.1546	0.1256	0.1207	0.1222
Negative Gamma jumps	1000	300	0.1905	0.1213	0.1205	0.0747
Negative Gamma jumps	2000	300	0.1973	0.0877	0.0889	0.0687
Negative Gamma jumps	3000	300	0.1978	0.0663	0.0871	0.0566

Table 2. Monte Carlo root-mean-square error of estimated regime tail indices, calculated against fixed generator values (q_0=1.12, q_1=1.42). Smaller is better.

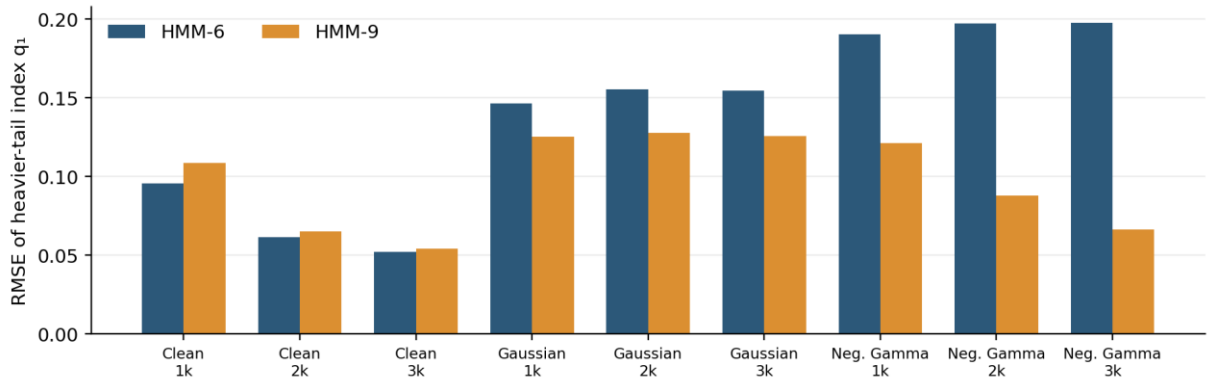


Figure 1. q_1 estimation RMSE by generator and sample size. HMM-9 is most useful under the correctly specified negative-jump generator; HMM-6 performs better for clean returns.

For negative Gamma jumps at n=3,000, HMM-9 reduces q_1 RMSE from 0.1978 to 0.0663, a 66.5% decrease. At n=1,000 under clean Student-t observations, the unnecessary jump extension raises q_1 RMSE from 0.0955 to 0.1088. The HMM-9 improvement is therefore concentrated where a correctly specified additional asymmetric-jump mechanism exists. Reductions under Gaussian contamination are not evidence that a negative-Gamma distribution correctly specifies symmetric contamination.

8.2 Filtered tail state and held-out density forecasting

Generator	n	Filter RMSE H6	Filter RMSE H9	Mean Δ log score	95% MC interval
Clean Student-t	1000	0.1180	0.1197	-0.00038	[-0.00059, -0.00018]
Clean Student-t	2000	0.1058	0.1060	-0.00008	[-0.00015, -0.00000]
Clean Student-t	3000	0.1012	0.1016	-0.00002	[-0.00007, +0.00003]
Gaussian contamination	1000	0.1874	0.1741	+0.00018	[-0.00027, +0.00062]
Gaussian contamination	2000	0.1787	0.1623	+0.00102	[+0.00059, +0.00145]
Gaussian contamination	3000	0.1779	0.1627	+0.00075	[+0.00036, +0.00115]
Negative Gamma jumps	1000	0.1948	0.1244	+0.01691	[+0.01551, +0.01832]
Negative Gamma jumps	2000	0.1900	0.1176	+0.01690	[+0.01565, +0.01815]
Negative Gamma jumps	3000	0.1881	0.1133	+0.01833	[+0.01716, +0.01950]

Table 3. Dynamic filtered-index RMSE and paired held-out one-step log predictive density score (HMM-9 − HMM-6). Positive gain favors HMM-9. Intervals refer to the mean across independent Monte Carlo paths; they are not parameter confidence intervals.

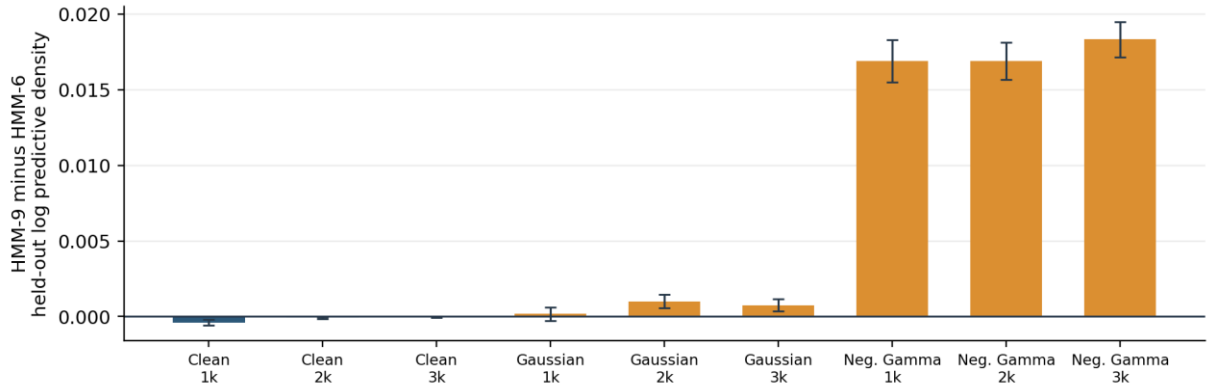


Figure 2. Paired predictive log-score improvement, with approximate 95% Monte Carlo intervals across independent paths. The primary density-score benefit is concentrated in the negative-Gamma generator.

In the negative Gamma generator at n=3,000, mean held-out log-score improvement is +0.01833 per observation (paired 95% Monte Carlo interval [0.01716,0.01950]); HMM-9 wins on 291 of 300 test paths. Gains at n=1,000 and n=2,000 are +0.01691 and +0.01690, respectively. Clean controls show approximately zero or slightly negative gains, while Gaussian contamination produces only much smaller changes. Monte Carlo intervals on forecast-score differences are not confidence intervals for model parameters.

8.3 Jump-parameter recovery and numerical stability

Generator	n	Mean $\hat{\varepsilon}$	Mean $\hat{k}$	Mean $\hat{\theta}$	Non-success H9	k at bound
Clean Student-t	1000	0.0028	1.805	3.160	1	14/300
Clean Student-t	2000	0.0009	1.671	3.216	0	3/300
Clean Student-t	3000	0.0020	1.716	3.204	0	4/300
Gaussian contamination	1000	0.0066	4.507	1.872	0	141/300
Gaussian contamination	2000	0.0069	4.844	1.565	1	159/300
Gaussian contamination	3000	0.0060	4.828	1.548	3	145/300
Negative Gamma jumps	1000	0.0321	3.207	2.670	1	53/300

Generator	n	Mean ε̂	Mean k̂	Mean θ̂	Non-success H9	k at bound
Negative Gamma jumps	2000	0.0321	2.498	2.984	0	18/300
Negative Gamma jumps	3000	0.0317	2.351	2.986	0	6/300

Table 4. HMM-9 jump parameter estimates and optimiser diagnostics. In the negative-jump generator the truth is (ε,k,θ)=(0.03,2,3); under clean data k and θ are unidentified at ε=0, so their means are not meaningful recovery statistics.

The balanced 2,700-path study has six unsuccessful HMM-9 optimiser-success flags and finite values in all recorded numerical result fields. Under negative-Gamma jumps at n=3,000, the mean estimates are $\hat{\varepsilon}$=0.0317, $\hat{k}$=2.351, and $\hat{\theta}$=2.986, against the generating values (0.03,2,3); only 6/300 k estimates touch the designated search boundary, compared with 53/300 at n=1,000. In contrast, under Gaussian contamination, the k boundary is hit in 145/300 at n=3,000, consistent with misspecification or weak identification. All such interpretation remains conditional on optimisation quality and discretisation accuracy.

9. Confidence-interval diagnostics for HMM-6 and HMM-9

The historical HMM-6 Wald experiment contains 500 independent datasets in each of four settings (clean and Gaussian contamination at n=1,000 or 3,000), or 2,000 paths total. Its observed-information Hessian is used for marginal 95% intervals and joint six-dimensional Wald statistics. The experiment is distinct from the matched forecasting evaluation. Results in Table 5 concern only HMM-6.

Generator	n	q_0 95% CI coverage	q_1 95% CI coverage	Joint Wald coverage	Valid Hessians / 500
Correct HMM-6	1,000	94.2%	94.6%	87.4%	500
Correct HMM-6	3,000	93.8%	93.6%	91.8%	500
Gaussian-contaminated	1,000	79.3%	39.7%	20.3%	479
Gaussian-contaminated	3,000	72.5%	5.6%	0.2%	499

Table 5. Independent historical 2,000-replication HMM-6 Wald study. Coverage conditions on positive-definite observed-information matrices, as specified in the source report.

Under correct six-parameter specification, marginal q_1 coverage is 94.6% (n=1,000) and 93.6% (n=3,000), whereas joint coverage is 87.4% and 91.8%, below the nominal 95%. Under Gaussian contamination, q_1 coverage falls sharply, illustrating how misspecification affects structural interval validity. These HMM-6 findings cannot establish HMM-9 coverage.

A separate nine-parameter Wald experiment can be made available for 500 negative-Gamma datasets at each of n=1,000 and n=3,000. Table 6 reports the fraction in which a full nine-parameter Hessian-based analysis was eligible, and conditional coverage given eligibility. Its conclusions are limited: a positive-definite interior Hessian is not obtained for every path, and the joint nine-dimensional coverage falls short of 95% even in the eligible subset. Earlier full-refit profile and parametric-bootstrap studies remain small pilots; reliable full-parameter bootstrap or profile-based coverage has not yet been established.

9.1 Nine-parameter Wald eligibility and coverage

The following summaries refer only to the completed HMM-9 negative-Gamma inference experiment. Coverage percentages are conditional on passing the observed-information eligibility rule, not unconditional 500-path coverage. An ineligible Hessian can result from boundary proximity, optimiser failure or non-positive-definite numerical curvature.

n	Datasets	Wald eligible	Eligible (%)	q_1 coverage	θ coverage	Joint 9D coverage
1000	500	287/500	57.4%	94.1%	84.3%	83.3%
3000	500	428/500	85.6%	93.9%	88.1%	89.7%

Table 6. HMM-9 observed-information diagnostics (500 negative-Gamma paths per training length). The coverage values condition on Wald eligibility; nominal target is 95%.

At n=1,000, only 287 of 500 fits are Wald eligible (57.4%); at n=3,000, eligibility improves to 428 of 500 (85.6%). Conditional coverage for q_1 is 94.1% and 93.9%, while conditional coverage for Gamma scale θ is only 84.3% and 88.1%. The conditional joint nine-parameter coverage is 83.3% and 89.7%. Hence this simulation does not validate routine nine-parameter Gaussian Wald inference; near-boundary failures and nonregular testing may require separate treatment.

10. Model comparison, causal interpretation, and robustness

The negative-Gamma extension is only appropriate when the data support an additional sign-asymmetric jump mechanism. A clean-data negative control is necessary because the augmented model can overfit observations already represented by Student-t tails. A symmetric Gaussian contamination experiment is also necessary because a reflected-Gamma innovation cannot represent arbitrary measurement noise; any gains in that scenario should be interpreted as partial robustness, not a correctly specified Gaussian model.

Robustness checks already explored in independent supplements include Isolation Forest anomaly moderation, sign-dependent and bounded cubic return signals, Gamma-precision EM, and alternative optimiser initialisations. Those experiments are informative but not equivalent to the nine-parameter likelihood test reported here. In particular, Isolation Forest training weights change the estimating objective and do not automatically yield valid Wald information; Gamma precision merely re-expresses the original Student-t law; and a predictive cubic coefficient belongs to a different hypothesis about conditional mean dynamics. None is silently added to Θ_9 in this manuscript.

Model choice should consider prequential predictive density scores, likelihood penalties, boundary rates, and sensitivity to plausible departures from the assumed generator. A separate progressive E1–E7 five-replication pilot indicates that volatility-estimation error can matter, but this small study is not pooled with the main Monte Carlo.

10.1 A theory-facing robustness requirement: location and asymmetry

In the reflected-Gamma generator, $E[r_t|K_t=j] = -\varepsilon k\theta = -0.18$. The centered HMM-6 lacks a location term; this could have exaggerated HMM-9's apparent advantage. We therefore fit a seven-parameter Student-t HMM with a shared estimated location μ using identical generated paths, the same chronological evaluation convention and an L-BFGS-B maximum-likelihood procedure. This controls directly for a constant conditional-mean shift while leaving the symmetric Student-t emission family intact. Proposition 3 predicts a qualitative difference between this location-only model and a genuine one-sided jump distribution.

10.2 Completed 2,700-path benchmark against a shared-location HMM

The location-adjusted estimator (HMM-7) augments Θ_6 with a common μ. It is fit by bounded L-BFGS-B using three numerical starts, including centred and sample-mean-shifted starting values; every reported location-model optimiser returned a successful flag. A known-parameter oracle is additionally evaluated using the true generator's emission law (Student-t, reflected-Gamma or Gaussian-convolution, respectively); it is an unattainable reference, not an estimated rival.

Generator	n	q1 RMSE H7	q1 RMSE H9	Δ log score	95% MC interval	H9 wins
Clean Student-t	1,000	0.0953	0.1088	+0.00007	[-0.00019, +0.00033]	158/300
Clean Student-t	2,000	0.0612	0.0651	+0.00020	[+0.00005, +0.00035]	162/300
Clean Student-t	3,000	0.0518	0.0543	+0.00008	[-0.00004, +0.00020]	152/300
Gaussian contamination	1,000	0.1469	0.1254	+0.00056	[+0.00006, +0.00105]	171/300
Gaussian contamination	2,000	0.1551	0.1279	+0.00150	[+0.00105, +0.00195]	200/300
Gaussian contamination	3,000	0.1544	0.1256	+0.00088	[+0.00048, +0.00129]	178/300

Negative Gamma jumps	1,000	0.1904	0.1213	+0.01709	[+0.01568, +0.01849]	273/300
Negative Gamma jumps	2,000	0.1972	0.0877	+0.01703	[+0.01578, +0.01828]	282/300
Negative Gamma jumps	3,000	0.1976	0.0663	+0.01830	[+0.01712, +0.01948]	291/300

Table 7. Matched HMM-7 (shared unknown location) versus HMM-9. q1 RMSE is measured against the generating q1=1.42. Positive log-score differences favor HMM-9. Intervals show variability of the mean paired score across independent simulated paths, not confidence intervals for structural parameters. Each cell has 300 paths.

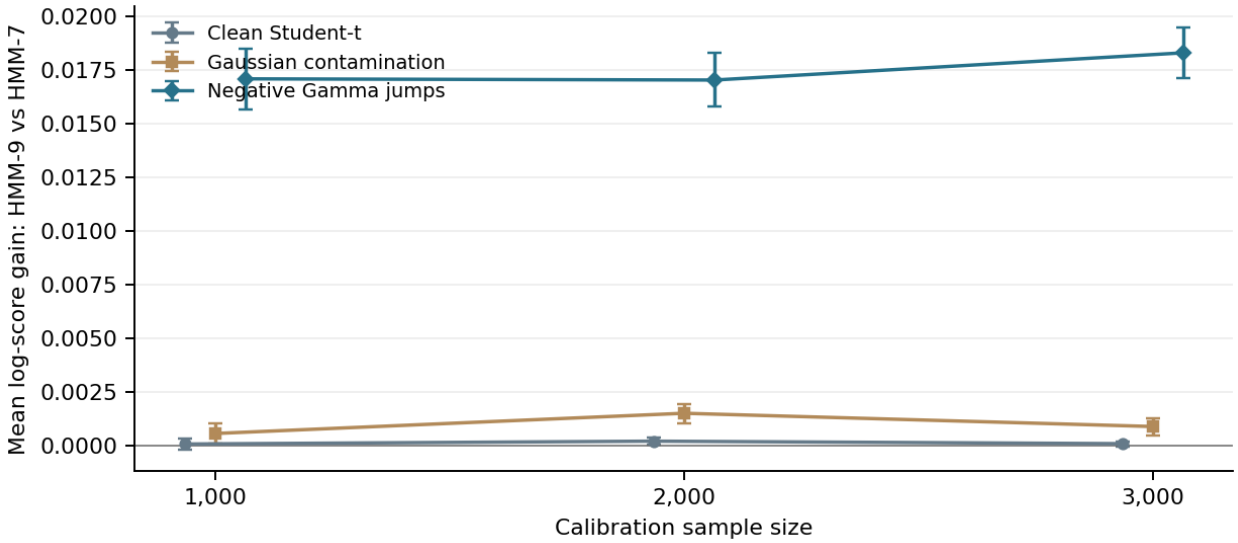


Figure 3 shows paired held-out predictive score improvement of HMM-9 over the location-adjusted HMM-7, with 95% Monte Carlo intervals across independent simulation paths.

At n=3,000 in the negative-Gamma generator, HMM-7's *q1* RMSE is 0.1976, while HMM-9's is 0.0663. The mean held-out score difference is +0.01830 per observation (paired 95% Monte Carlo interval [0.01712, 0.01948]), and HMM-9 wins on 291 of 300 paired paths. The estimated HMM-7 location averages about −0.0207, much smaller than the generating unconditional mean shift of −0.18. This is consistent with a symmetric Student-t likelihood that accommodates a few large negative jumps through tail or scale adjustments rather than a global shift; it is not, by itself, proof of structural parameter identification.

Under clean Student-t observations, the log-score differences between HMM-9 and HMM-7 are close to zero, while the simpler models typically estimate *q1* more precisely. Under symmetric Gaussian contamination, HMM-9 shows smaller but sometimes positive gains, even though its one-sided Gamma component is misspecified. Therefore, the results support the substantive distinction between tail thickness, location, and asymmetric jumps in the correctly specified design, without showing that HMM-9 is superior to every possible skewed Student-t or general contamination model.

10.3 Oracle forecasting and remaining econometric inference

Generator at n=3,000	HMM-7	HMM-9	True-generator oracle
Clean Student-t	-1.46717	-1.46709	-1.46614
Gaussian contamination	-1.58342	-1.58254	-1.57722
Negative Gamma jumps	-1.62356	-1.60526	-1.60372

Table 8 shows mean held-out per-observation log predictive density. The oracle knows the true data-generating parameters and uses the appropriate emission family, including the symmetric Gaussian-contamination convolution. Higher scores are better.

The oracle log scores provide a ceiling only in expected log-score terms, not necessarily for every finite realized test sample. HMM-9 comes substantially closer to the correctly specified negative-Gamma oracle than HMM-7 does. The experiment does not validate structural Wald intervals or prove a full nine-parameter identification theorem. Model-family non-equivalence in Proposition 3 is a weaker and more targeted theoretical statement than

unique recovery of θ_9; both must be distinguished. The independent 500-path-per-sample HMM-9 inference experiment in Section 9 continues to govern confidence-interval qualifications.

The comparison remains simulation-based. Validation on independently held-out real observations, and benchmarking against flexible skew-t and other asymmetric-jump or contamination families, would clarify external relevance. Those experiments are not necessary to define the present model, but the current Monte Carlo findings alone cannot establish superiority for actual financial or physical series.

11. Reproducibility and data provenance

The principal Monte Carlo dataset is all_replications_combined_v2.csv, with 2,700 unique (scenario, n, rep) records, 300 paths in each of nine scenario/training-length cells, and no missing numerical entries. Results were audited against the original cell files, including the completed correct_n2000_reps0-299.csv. The enlarged design shares simulation identifiers and random-seed conventions with earlier studies; neither the older 500-path benchmark nor any earlier partial checkpoint is pooled as an independent sample in Tables 1–4. The completed shared-location extension is archived as GT_Location_Complete_2700_Replications.csv and GT_Location_Complete_2700_Summary.csv, generated by gt_hmm7_location_benchmark.py using the same source generator, ids, and fixed seed logic. The known-parameter Gaussian-contamination oracle uses grid convolution cross-checked against numerical integration.

The associated source code is giorno_tsallis_complete.py. Its FFT-based convolution and optimization settings should be archived with exact dependency versions and run commands. The independent HMM-9 Wald evidence is drawn from inference_replications.csv and inference_coverage_summary.csv (500 paths each for n=1,000 and 3,000). HMM-6 Wald statistics come from the separate Giorno_Tsallis_P12_Large_Wald_Study study. The E1–E7 dynamic-volatility experiments are exploratory pilots and remain separate. None of these experiments includes a real-market evaluation.

12. Conclusion

The completed 2,700-path paired Monte Carlo study supports a theory-driven empirical claim: when observations include independent negative Gamma jumps, an explicit asymmetric convolution improves latent tail-index recovery and predictive density forecasts relative not only to a centred HMM-6 but also to a newly estimated shared-location HMM-7. At n=3,000, q1 RMSE is 0.1978 (HMM-6), 0.1976 (HMM-7) and 0.0663 (HMM-9). Relative to HMM-7, the held-out mean log-score gain is 0.01830 per observation, with a paired 95% Monte Carlo interval [0.01712, 0.01948]. Proposition 3 supplies an exact, limited theoretical foundation for why a nondegenerate reflected-Gamma mechanism cannot be reduced to a common location shift, independently of whether third moments exist. This does not imply superiority over every alternative skewed model, nor does it prove global identification of all nine parameters. It only points a direction for future improvements.

The HMM-9 Wald study adds a more qualified finding: although eligible interior fits yield near-nominal marginal coverage for some parameters, only 57.4% and 85.6% of fits at n=1,000 and n=3,000 pass its Hessian-eligibility rule, and conditional nine-dimensional joint coverage remains below nominal. Thus neither forecast accuracy nor Hessian positivity proves global identification or calibrated inference. Structural breaks and time-varying Gamma shape are deliberately outside the scope of the constant-parameter HMM-9 study.

12.1 Future research: volatility-dependent Gamma shape

A natural extension of the constant-k HMM-9 is to permit the Gamma shape to respond to an observed or filtered conditional-volatility process. A candidate specification proposed for subsequent study is

$$k_t = k_0 \exp[\eta \log(\frac{\sigma_t}{\sigma_{ref}})], \quad k_0 > 0, \quad \sigma_t > 0, \qquad (11)$$

where σ_ref is a fixed reference-volatility level, η is the volatility elasticity of Gamma shape, and η=0 nests the time-invariant HMM-9 shape. In preliminary five-replication progressive E1–E7 experiments, extensions incorporating volatility and conditional-mean forecasting did not demonstrate stable recovery of η, even when true conditional volatility and mean were supplied to the estimator. This is suggestive of weak joint identification or challenging likelihood geometry—not a theorem of nonidentification. Future work should profile η with k_0, jump probability ε and scale θ sequentially released, establish conditions on volatility variation and jump frequency, and validate uncertainty by full-refit simulation before interpreting a fitted k_t as structurally identified.

12.2 Future research: structural breaks, change points, and parameter stability

The current HMM admits recurring changes in the unobserved regime K_t, but assumes a fixed transition matrix P and fixed observation parameters Θ_9 over the estimation sample. A latent state transition under an unchanged model is different from a structural break, whereas a break changes the mechanism governing transitions, tail indices, scales, jump incidence, or jump magnitudes. Market disruptions may therefore generate episodes that a constant-parameter HMM mistakenly interprets as prolonged high-tail-risk states or recurrent negative jumps.

A parsimonious future extension would permit piecewise-constant parameters $\Theta_{9,t} = \Theta_9^{(m)}$ *for* $\tau_{m-1} < t \leq \tau_m$, with change points τ_m unknown, while retaining the normalised q-Gaussian–Gamma emission law within each segment. Another approach would allow selected parameters—especially p_{00}, p_{11}, ε, and the Gamma scale θ—to evolve through a separate latent structural-break state. Such models require explicit identification restrictions: an apparent break in ε or θ may otherwise be confounded with changes in q, s, or latent-state persistence.

The corresponding research programme should compare no-break, single-break, multiple-break, and smoothly evolving specifications on paired simulations; report detection delay and false-alarm frequency as well as predictive scores and parameter recovery; and examine inference when break dates are estimated rather than known. In particular, conventional fixed-parameter Hessian tests and standard χ^2 reference distributions need not remain valid under unknown break points or weakly identified jump parameters. Break models and dynamic k_t should first be estimated separately and only then combined, to avoid obscuring which source of time variation the observed data can identify.

A further cross-disciplinary direction is to assess whether latent q-Gaussian backgrounds contaminated by intermittent asymmetric impulses can be distinguished in well-characterized physical observations. This would involve system-specific sampling and calibration, sign-sensitive residual diagnostics, and comparisons with established physical noise models. The present statistical model should not be interpreted as deriving nonextensive thermodynamics, nor as establishing an entropy-production law.

12.3 Future research: cross-model comparison of latent q and entropy-oriented estimators

A natural next study will compare HMM-9 with two distinct estimation strategies rather than treating all quantities labeled q as identical. First, for a common time series, estimate latent regime-specific q values using HMM-9 and evaluate competing heavy-tail/impulse families, including nonparametric or window-specific approaches where justified. Second, evaluate a common Tsallis entropy functional S_a of the models' normalized one-step predictive distributions for a prespecified entropy order a. The benchmark must disentangle improvements due to latent-state filtering from those due to allowing negative shifts, skewness, time-varying scale, or jumps.

Such comparisons could be designed with matched simulations that vary tail regimes, impulse frequency, amplitude and sign, plus a null with no extra shocks. Methods based on seismological q-exponential relations, class-specific adaptive q, or quantum-relative-entropy estimation cannot simply be inserted as numerically equivalent estimators of our q_j: they answer different questions. Instead they inform alternative measurement designs and the scientific distinction between determining q and measuring an entropy functional. Report bias, RMSE, predictive scores, model-selection behavior, robustness to misspecification, and the reliability of inferred q changes.

12.4 Future research: independent validation of predictive Tsallis entropy

Define an entropy order a>0 with a≠1 before examining the data, use a fixed dimensionless transformation x of the observations, and let p̂_t(x) denote the normalized HMM-9 one-step predictive density. A plug-in predictive entropy functional can then be computed, subject to existence of the integral, as:

$$\hat{S}_{a,t} = [1 - \int \hat{p}_t(x)^a \, dx] / (a - 1).$$

This is entropy of the complete predictive mixture, not the posterior-weighted mean of regime-specific entropies and not an entropy formula obtained by substituting q̂_t into a single q-Gaussian density. Compare this plug-in functional with independently estimated Tsallis entropies from sample-spacing and quantile methods, including the approaches studied by Chakraborty, Nanda, and Balakrishnan (2026). Assess bias, interval coverage, finite-sample variability, sensitivity to the order a, contamination and rare jumps, and the behavior as a approaches 1 (the Shannon limit). Direct estimators may need adaptation to dependent, nonstationary data: their i.i.d. regularity conditions cannot be assumed to transfer without proof.

The principal validation question is whether the fitted latent tail index q̂_t, the corresponding predictive distribution, and the independently estimated entropy functionals provide mutually consistent evidence when each is interpreted on its own scale. Agreement could support statistical calibration; disagreement would motivate diagnostics for model misspecification, scale handling and temporal dependence. Neither outcome alone identifies a universal physical q.

12.5 Future research: physical transferability and testing the universality conjecture

A separate physical application could test whether a hidden q-Gaussian background plus an intermittent impulse model improves state recovery in a system for which an observation mechanism can be defended. Seismic catalogs are one possible starting point because the reviewed literature estimates indices associated with magnitude, inter-event time and distance. However, our additive, negatively signed Gamma disturbance was constructed for financial returns; its sign, support, scale and occurrence process must be re-specified rather than transferred literally to earthquake magnitudes, spin lattices or quantum states.

A progressive experiment should (i) choose a physically meaningful observable and dimensionless reference scale; (ii) compare static q fits with latent-regime models and physically motivated background/impulse alternatives; (iii) validate inferred states against independent physical diagnostics; (iv) investigate changes in q around interventions, transitions or externally verified events; and (v) test whether inferred relationships replicate across datasets and measurement protocols. In spin-lattice applications, independently observable equilibrium diagnostics would be needed to test any proposed relation between q and distance from equilibrium; in quantum systems the relative-entropy estimation problem remains mathematically different.

The long-term scientific aspiration is to search for an entropy framework capable of accounting for regularities across different classes of physical and economic systems. This aspiration is not a claim that one universal entropy, one numerical q, or one HMM-9 mechanism already explains them all. Establishing any common principle would require cross-system predictions, falsifiable restrictions and evidence beyond the current financial Monte Carlo study.

Appendix A. Mathematical notes and proof sketches

A.1 Student-t/q-Gaussian identification

Take two centered densities $f(\cdot; \nu, s)$ and $f(\cdot; \nu', s')$ of equation (2) agreeing almost everywhere, and use continuity to obtain equality pointwise. As $|r|\to\infty$, $\log f(r; \nu, s)/\log |r| \to -(\nu+1)$. Equality forces $\nu=\nu'$. At $r=0$, $f(0; \nu, s)=c(\nu)/s$ with $c(\nu)>0$, so $s=s'$. Since $q = \frac{\nu+3}{\nu+1}$, $\nu > 0$, is strictly decreasing for $\nu>0$ or $\frac{dq}{d\nu} = \frac{-2}{(\nu+1)^2} < 0 \quad (\nu > 0)$, the mapping to q is injective. This proves the emission-family statement only.

A.2 Proper normalization and Bayes recursion

For any state density f and independent nonnegative Gamma G, $\int\mathbb{R}$ f(r+z)dr=1 for fixed z. Tonelli's theorem gives $\int\mathbb{R}\int_0\infty$ f(r+z)gΓ(z)dz dr=1; hence equation (6) is normalised. Conditional prediction follows the finite-state Chapman–Kolmogorov equation, and updating with g yields equation (8). No bounded-support assumption is needed for this formal recursion.

A.3 Why the no-jump null is nonregular

At $\varepsilon=0$, $\partial g_j/\partial k$ and $\partial g_j/\partial\theta$ vanish because the convolution term is multiplied by ε; the data likelihood is invariant to k and θ. Thus, the full nine-parameter Fisher information cannot be nonsingular under the null, and routine interior Wald/χ^2 asymptotics fail. Testing jump existence requires a carefully constructed nonstandard or bootstrap procedure, with the bootstrap itself validated under the boundary null.[8]

A.4 The forecast score does not prove parameter inference

An average held-out log-score gain is a proper scoring-rule comparison for predictive distributions under a defined data generator, whereas a 95% confidence interval for q_1 concerns an estimator's sampling distribution around the true parameter. They are logically different propositions. The reported Monte Carlo confidence intervals for mean score improvements cannot be substituted for parameter-coverage checks.

A.5 Proof of Proposition 3: marginal asymmetry cannot be a global location shift

For an arbitrary time t, write $X_t = s_{K_t}T_{\nu_{K_t},t}$. Conditional on $K_t=j$, the Student-t law can be represented by $X_t \mid \Lambda_t, K_t = j \sim N(0, \frac{{s_j}^2}{\Lambda_t})$, with a strictly positive Gamma mixing variable Λ_t. Hence its characteristic function $\varphi_j(u) = E[exp(\frac{-u^2{s_j}^2}{2\Lambda_t})]$ is real, even and strictly positive at every real u. Mixture over the contemporaneous probabilities $Pr(K_t=j)$ yields $\varphi_X(u) = \sum_j Pr(K_t = j)\varphi_j(u) > 0$, real and even, irrespective of the initial state distribution.

Let $J_t = -B_tG_t$, independent of X_t, where $Pr(B_t=1) = \varepsilon>0$ and $G_t \sim Gamma(k,\theta)$, $k, \theta>0$. If $Y_t = X_t + J_t$ were symmetric about some finite c, then $e^{-iuc}\varphi_Y(u)$ would be real and even. Since $\varphi_Y(u) = \varphi_X(u)\varphi_J(u)$ and $\varphi_X(u) > 0$ is real and even, $e^{-iuc}\varphi_J(u)$ must likewise be real and even. By characteristic-function uniqueness, J_t-c would be symmetric about zero. This is impossible: J_t is bounded above by zero, but, because a Gamma distribution has unbounded positive support and $\varepsilon>0$, J_t is unbounded below. No probability law with one-sided bounded support and an unbounded tail in the opposite direction can be centrally symmetric about any finite location.

Every shared-location HMM-7 marginal instead has form $\mu + \tilde{X}_t$ where $\tilde{X}_t$ is a mixture of centred Student-t emissions and is symmetric about zero. Consequently a non-degenerate HMM-9 marginal cannot equal any shared-location HMM-7 marginal. QED. Scope: this establishes one-time marginal distribution-family non-equivalence (and therefore separation of these two model classes), not injectivity of the map $\Theta_9 \mapsto law(r1 : T)$, uniqueness against arbitrary skewed-innovation models, or nominal nine-parameter inference.

Appendix B. Boundary nesting, Shannon limit, and jump-model inference

The results below concern the observational models of paper. They do not establish a limit of CARA optimal controls, which belongs to another future paper. In particular, Tsallis entropy (1988, 2009) is a statistical description here, not the economic utility functional.

[8] Hansen, B. E. (1992)

Lemma B.1 (Gaussian limit of the q-Gaussian/Student-t emissions)

Let $\nu(q) = \frac{3-q}{q-1}$, $q \in (1,5/3)$, and let $f(r; q, s)$ be a centered Student-t density with $\nu(q)$ degrees of freedom and Student-t scale $s>0$. If $q_m \downarrow 1$ and $s_m \to s_0 \in (0, \infty)$, then $f(\cdot; q_m, s_m)$ converges in distribution and in L^1 to the $N(0, s_0^2)$ density. This is a limit of densities; the Gaussian lies on the $q=1$ boundary when the estimated parameter set excludes $q=1$.

$$\nu(q) = \frac{3-q}{q-1} \to \infty \text{ as } q \downarrow 1. \qquad \text{(B.1)}$$

Proof. Write $Tv = Z / \sqrt{(Uv/v)}$, where Z is standard normal, Uv is independent chi-square with v degrees of freedom, and Tv is standard Student-t. Since $Uv/v \to 1$ in probability as $v \to \infty$, Slutsky's theorem gives $s_m Tv(q_m) \Rightarrow s_0 Z$. Student-t densities converge pointwise to the $N(0, s_0^2)$ density; all are normalised, hence Scheffé's lemma yields L^1 convergence. QED.

Lemma B.2 (Tsallis–Shannon entropy limit)

For a finite probability vector $p=(p_1,\ldots,p_m)$, with $0 \log 0$ defined as zero, let $S_q(p)=[1-\sum_i p_i^q]/(q-1)$. Then $S_q(p) \to -\sum_i p_i \log p_i$ as $q \to 1$. For continuous distributions, interchange of the limit and integration requires additional domination/integrability conditions.

$$\lim_{q \to 1} S_q(p) = -\sum_i p_i \log p_i \qquad \text{(B.2)}$$

Proof. The numerator and denominator both vanish at q=1. Differentiation in q gives $-\sum_i p_i^q \log p_i$; L'Hôpital's rule gives the stated limit. Terms with $p_i=0$ contribute zero by continuity. QED.

Corollary B.3 - Nested models and the non-regular no-jump boundary.

Consider the nine-parameter state-j emission density $g_j(r) = (1-\varepsilon) f_j(r) + \varepsilon \int_0^\infty f_j(r+z)\, g_\Gamma(z; k, \theta)\, dz$. (a) At $\varepsilon=0$ this reduces exactly to the six-parameter q-Gaussian HMM observation likelihood, whatever k and θ are. (b) If $\varepsilon \to 0$, $q_0, q_1 \to 1$ and $s_0, s_1 \to s>0$, both emission densities converge to the same Gaussian density. The observed return model then reduces to a one-component Gaussian observation law when the hidden state influences observations only through these emissions.

$$g_j(r) \to \varphi(r; 0, s^2) \text{ if } \varepsilon \to 0,\ q_j \to 1,\ s_j \to s. \qquad \text{(B.3)}$$

Proof. For (a), substitute $\varepsilon=0$. For (b), the L^1 distance between g_j and f_j is bounded by 2ε, because both f_j and the convolution are probability densities. By Lemma B.1 the Student-t component approaches φ. The triangle inequality gives L^1 convergence. When the emissions are identical across states, the marginal observation density no longer depends on the latent transition parameters. QED.

Identification warning. At $\varepsilon=0$, the Gamma shape k and scale θ disappear from the likelihood and are unidentified. If $q_0=q_1=1$ and $s_0=s_1$, the two latent states are observationally indistinguishable and transition probabilities are unidentified. Therefore, this is a boundary/closure nesting result, not a regular nested-model likelihood-ratio or nine-parameter Wald theorem. The strict ordering convention $q_0<q_1$ also has to be relaxed in the closure.

Remark B.4 -What the nesting does not prove.

Although the q-Gaussian observation model converges to Gaussian, and Tsallis entropy converges to Shannon entropy, this does not make CARA market-making expected utility converge by the same identity: the objective and action constraints must also be specified and shown to converge. A jump model with fixed $\varepsilon>0$ does not converge to a pure Gaussian emission simply because $q \to 1$; it converges to a Gaussian–negative-Gamma mixture.

Implication for the nine-parameter identification program

The preceding results justify treating the no-jump and Gaussian cases as informative negative controls, while showing why conventional tests of $\varepsilon=0$ cannot use a standard χ^2 reference distribution without further analysis. A publishable nine-parameter identification theorem must be established on an appropriate interior parameter set with $\varepsilon>0$,

separated state emissions, a full-rank transition matrix, regular quadrature and a non-singular information matrix; these are proposed sufficient-condition checks, not already-proven global identification for the convolution HMM.

Research boundary. The proposed finite-horizon belief-state regret theorem and q-dependent inventory-risk derivative were checked separately and added only to the future paper. This separation preserves the econometric emphasis of this paper.

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